M101 · Unit 12 · Quick reference

Validation, sensitivity and competing answers
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Recall **Compare like with like**	Quantity, units, interval, initial value, model and inputs.
Recall **Discrepancy**	Signed: $Q - I$. Absolute: $\lvert Q - I \rvert$. Relative: $\lvert Q - I \rvert$ / $\lvert I \rvert$, only $I \neq 0$.
Understand **Accumulation**	$A(T) = \int_0^T q(t)\,dt$; check $A(0) = 0$ and $A'(T) = q(T)$ for continuous q. Stored amount = initial amount + net accumulation.
Understand **Local sensitivity**	$\Delta F = F(p + \delta p) - F(p) \approx F'(p)\delta p$. Small admissible changes; check accuracy over the chosen interval.
Consult **Pump model**	$q(t) = a - bt^2$; $A(T) = aT - bT^3$ / 3. Fixed T: $\Delta A = T\delta a - T^3\delta b$ / 3. Check domain, rate sign and capacity.
Understand **Input ranges**	Find extrema over allowed inputs; monotonicity can locate endpoint/corner values. Bounds are not probabilities.

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Understand **Residual**	Residual = left side − right side. Small residual need not mean small solution error in a sensitive system.
Consult **Numerical comparison**	$\lvert u - v \rvert \leq \max(\varepsilon_{abs}, \varepsilon_{rel} \max(\lvert u \rvert, \lvert v \rvert))$. Absolute tolerance has output units; relative tolerance is dimensionless.
Consult **Quadratic trapezoid error**	For $q = a - bt^2$ on [0,T], n equal panels: $Q_n - A(T) = -bT^3 / (6n^2)$. Exact arithmetic; fixed model/inputs.
Understand **Independent evidence**	Constant/shifted-linear benchmarks; reversed/zero interval; known curved case; domain and samples; fresh run.
Consult **Diagnose**	Separate model, input, approximation, implementation and rounding. Vary one factor; retain precision until display.
Consult **Report**	Question → model/domain → method → independent checks → discrepancy/tolerance → limitations. State assistance used.

Consult · Python

```
from math import isclose
# Values expressed in the same units:
isclose(u, v, rel_tol=0.001, abs_tol=0.0005)
# Signed/absolute discrepancy:
difference = estimate - benchmark
error = abs(difference)
# Sensitivity with other inputs fixed:
change = F(p + dp) - F(p)
# Keep precision; format only for display:
print(f"{estimate:.6f}")
```

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